

LECTURE #1

Econometrics I

INTRODUCTION TO ECONOMETRICS & COURSE INFORMATION & QUICK STATISTICS REVIEW

Jiri Kukacka, Ph.D.

Institute of Economic Studies, Faculty of Social Sciences, Charles University

Summer semester 2026, February 17

Outline

Introduction to econometrics

Course information

Quick statistics review

- Probability distributions and their features

- Recap of selected distributions

- LLN and CLT

- Hypothesis testing

Outline

Introduction to econometrics

Course information

Quick statistics review

- Probability distributions and their features

- Recap of selected distributions

- LLN and CLT

- Hypothesis testing

What is econometrics?

- ▶ Econometrics has evolved from statistics as a separate discipline:
 - ▶ experimental vs. non-experimental (observational) data,
 - ▶ analysis of non-experimental data comes with additional issues.
- ▶ Focus on:
 - ▶ estimating economic relationships,
 - ▶ empirical testing of economic theories or hypotheses,
 - ▶ evaluating economic policies,
 - ▶ predicting, forecasting economic variables.
- ▶ Readings:
 - ▶ Chapter 1: 1.1, 1.2, 1.4.

Econometrics as a separate discipline

- ▶ 1930: the Econometrics Society in Cleveland, OH, USA
 - ▶ Joseph A. Schumpeter, Harold Hotelling, Karl Menger, Norbert Wiener, or Irving Fisher (president), among others.
 - ▶ 1932: the Cowles Commission for Research in Economics, Colorado Springs, CO, USA (due to Alfred Cowles)
 - ▶ Nobel Prize laureates: e.g., Keneth Arrow, Gérard Debreu, James Tobin, Franco Modigliani, Herbert A. Simon, Joseph E. Stiglitz, or Harry Markowitz.
- ▶ Alfred Cowles
- ▶ Many Nobel Prize laureates in economics are, in fact, econometricians
 - ▶ e.g., Robert Merton, Myron Scholes, James Heckman, Daniel McFadden, Robert Engle, Clive Granger, Thomas Sargent, Christopher Sims... 2021: Joshua Angrist, Guido Imbens.

Steps in empirical economic analysis

1. Proposing an economic 'model in words',
2. Converting it into a formalized econometric model,
3. Collecting data,
4. Selecting software/implementing code,
5. Estimating the model, evaluating its quality,
6. Reporting and interpreting the results.

Economic vs. econometric models

- ▶ Economic models are based on economic theory.
- ▶ Usually, only a general form of relationships is given:

$$y = f(\bullet).$$

- ▶ Econometric models have a specific form that can be estimated and further analyzed:

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k + u,$$

$$y = \beta_0 x_1^{\beta_1} \dots x_k^{\beta_k} u,$$

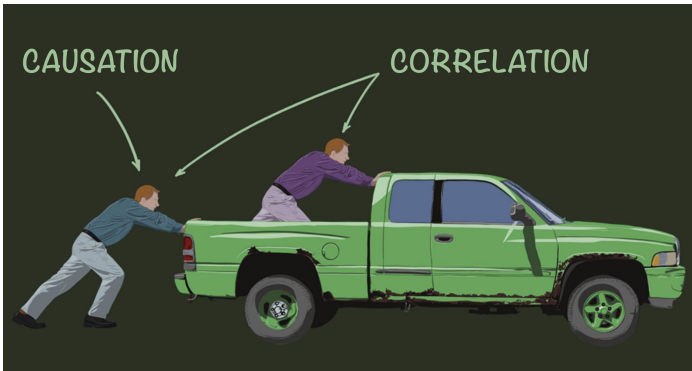
$$\vdots$$

with parameters $\beta_0, \dots, \beta_k$ and an error term u .

Note on causality and ceteris paribus in econometrics

- ▶ In an ideal case, the econometric analysis leads to uncovering a causal effect between variables.
- ▶ However, as the saying goes, “correlation is not causation”.
- ▶ The ceteris paribus approach (‘holding all other relevant factors fixed’) utilized in the econometric analysis could help with the issue $\Rightarrow$ econometric methods can simulate a ceteris paribus experiment.
- ▶ Nevertheless, finding a true causal effect is challenging.

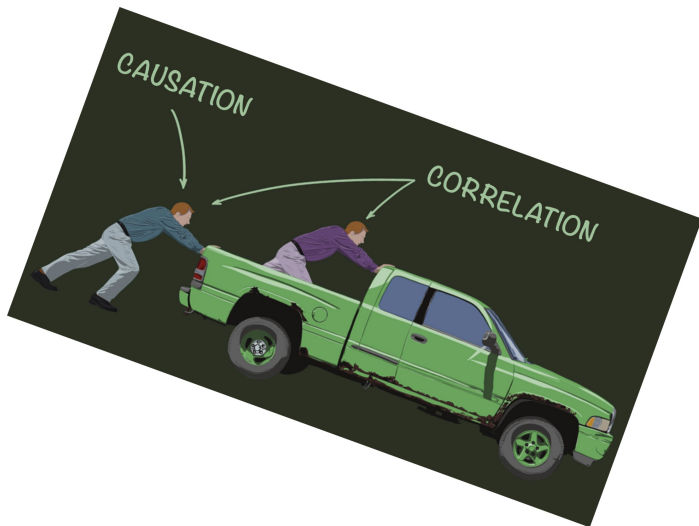
Causality in econometrics



Source: facebook.com/VedatorCZ [2020-10]

► Further examples

Ceteris paribus in econometrics



Source: facebook.com/VedatorCZ [2020-10]

Outline

Introduction to econometrics

Course information

Quick statistics review

- Probability distributions and their features

- Recap of selected distributions

- LLN and CLT

- Hypothesis testing

Econometrics I: People and times

- ▶ **Lecturer:** Jiri Kukacka, Ph.D.
 - ▶ email: jiri.kukacka@fsv.cuni.cz
 - ▶ web: ies.fsv.cuni.cz/en/contacts/institute-members/58305408
- ▶ **Code:** [JEB109](#) (link to SIS).
- ▶ **Lectures:** Tuesday, 11:00, lecture hall 'Michal Mejstrik' 109.
- ▶ **Office/consultation hours:** Tuesday, 13:30–15:30, room 406, by appointment via email, or we can schedule a Meet call.
- ▶ **Seminars:** Tuesday 12:30 [T] and 15:30 [H], Wednesday 15:30 [I] and 18:30 [M], PC lab 016.
- ▶ **Teaching assistants:**
 - ▶ Mgr. Tomas Karhanek: 47860177@fsv.cuni.cz
 - ▶ Mgr. Matej Korinek, MSc.: matej.korinek@fsv.cuni.cz
 - ▶ Hieu Nguyen, M.A.: HieuThiHoang.Nguyen@cerge-ei.cz
 - ▶ Mgr. Ivan Trubelik: ivan.trubelik@fsv.cuni.cz

Econometrics I: Requirements and grading

- ▶ **Pre-requidity:** [JEB105 Statistics](#) ' & ' Mathematics I, II, (III).
- ▶ **Course requirements:**
 - ▶ 2 home assignments (0–20 points: 10 + 10),
 - ▶ midterm written test (0–30 points),
 - ▶ final written exam (0–50 points, necessary condition for passing: more than 25 points),
 - ▶ detailed info in SIS: [JEB109](#).
- ▶ **Standard FSV grading policy ([Directive S_SO_002](#)):**
 - ▶ 90+ to 100 points result in 'A' ('Excellent'),
 - ▶ ...
 - ▶ 50 or fewer points result in 'F' (not passed),
 - ▶ partial activities started but final not attended $\Rightarrow$ 'F'!
- ▶ **6 (ETCS) credits:** equivalent to [150–180 hours](#) of study.

Econometrics I: Details and dates

► Home assignments:

- announced via SIS on Wed/Thu, due in 12 days,
- teams of two, completely via SIS ('Study group roster' app),
- 'academic integrity', deadlines non-negotiable, solo $\Rightarrow$ 0,
- points and feedback via SIS.

► Midterm test:

- 65-minute, written, closed-book, IDs, MM 109,
- **March 31**, 11:00, results via SIS.

► Final exam:

- 100-minute, written, closed-book, IDs, MM 109,
- exam terms (always from 11:00): **May 26, June 2, June 9, June 18 (Thu) Xor June 29 (Mon)**,
- resit term: **September 1** (11:00),
- registration (from May 4, 8:00 a.m., **capacity limited**) and results via SIS,
- specific details (changes) at the beginning of May.

Econometrics I: Materials and software

► **Course materials:**

- lecture handouts and R codes (Sun),
- seminar handouts, datasets, and R codes (Sun),
- a sketch of solutions and answers to exercises (Wed/Thu),
- solutions to home assignments (Tue),
- a specimen final exam (beginning of May).

► **Software:**

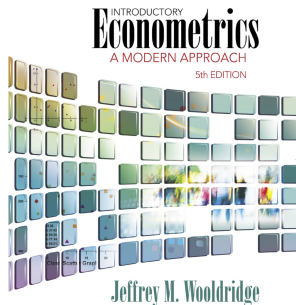
- [R](#) and [RStudio](#) (freeware, all platforms, PC lab 016),
- mostly at seminars (+ 'pen and paper' + discussion), BYOD?
- strongly suggested course: [Data Analysis in R](#) (if you are new to R, review at least the recorded introductory lectures).

► **Communication, please, in advance:**

- Q $\longrightarrow$ SIS: [JEB109](#) $\longrightarrow$ TAs $\longrightarrow$ lecturer/consultation $\longrightarrow$ A.

Econometrics I: Core textbook

Wooldridge, J. M. (2012). *Introductory Econometrics: A Modern Approach*. Cengage Learning, 5th Ed., .pdf (6th, [7th](#), [8th Ed.](#)).



Alternative (more advanced):

- ▶ [Baltagi, B. H. \(2021\)](#). *Econometrics*. Springer, 6th Ed.
- ▶ [Greene, W. H. \(2018\)](#). *Econometric Analysis*. Pearson Education, 8th Ed.
- ▶ [Wooldridge, J. M. \(2010\)](#). *Econometric Analysis* ... MIT Press, 2nd Ed.

Econometrics I: Syllabus & schedule

- ▶ L #1: Introduction & Information & Quick stats review
 - ▶ no seminars this week.
- ▶ L #2 – #4: Chapters 1 + 2 + 3, HA #1 due on March 23
 - ▶ *Simple + Multiple regression analysis: Basics + OLS.*
- ▶ L #5 and #6: Chapters 4 + 5
 - ▶ *Gauss-Markov theorem + Inference + Asymptotics.*
- ▶ Midterm test instead of lecture: March 31
 - ▶ no seminars this week.
- ▶ L #7 – #9: Chapters 6 + 7, HA #2 due on May 4
 - ▶ *Further issues + Selection of variables + Qualitative analysis.*
- ▶ L #10 and #11: Chapters 8 + 9
 - ▶ *Heteroskedasticity + Specification and data issues.*
- ▶ L #12: Revision of key concepts (Chapters 1–9)
 - ▶ no seminars this week.
- ▶ detailed info in SIS: [JEB109](#)

Outline

Introduction to econometrics

Course information

Quick statistics review

- Probability distributions and their features

- Recap of selected distributions

- LLN and CLT

- Hypothesis testing

Quick statistics review

- ▶ We will now recap the essential concepts students should be familiar with from their statistics courses that will be used throughout our econometrics classes.
- ▶ It will be a very brief, quick, and intuitive overview, primarily for information on what students are expected to know.
- ▶ Please refer to [JEB105 Statistics](#) and the Wooldridge (2012) appendices for details.
- ▶ Readings:
 - ▶ SIS information: [JEB109](#),
 - ▶ Appendix B, C.3, C.6.

Outline

Introduction to econometrics

Course information

Quick statistics review

- Probability distributions and their features

- Recap of selected distributions

- LLN and CLT

- Hypothesis testing

Distribution functions

- ▶ We deal with random variables that form a random sample (*i.i.d* n -tuple, usually with unknown probability distribution) and their realizations (observations $\rightarrow$ observed data set).
- ▶ For any random variable X , the cumulative distribution function (CDF) F_X is defined as

$$F_X(x) = P(X \leq x), \quad x \in \mathbb{R}.$$

- ▶ Some of the properties of F_X :
 - ▶ nondecreasing,
 - ▶ $\lim_{x \rightarrow -\infty} F_X(x) = 0$,
 - ▶ $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
- ▶ For continuous random variables, we have the density function $f_X(x)$ of X such that

$$F_X(x) = \int_{-\infty}^x f_X(t) dt, \quad -\infty < x < +\infty.$$

Selected moments and other features

- ▶ Mean and its estimators: sample mean, median,
- ▶ Variance and standard deviation,
- ▶ Bivariate association: covariance and correlation.

(Population) mean

- ▶ Measure of central tendency of a random variable.
- ▶ A.k.a. expected value $\mathbb{E}(X) = \int_{-\infty}^{+\infty} xf_X(x)dx$.
- ▶ Some of the properties:
 - ▶ $\mathbb{E}(X + c) = \mathbb{E}(X) + c$, where c is a constant,
 - ▶ $\mathbb{E}(X + Y) = \mathbb{E}(X) + \mathbb{E}(Y)$,
 - ▶ $\mathbb{E}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \mathbb{E}(X_i)$,
 - ▶ $\mathbb{E}(cX) = c\mathbb{E}(X)$, where c is a constant.
- ▶ Popular estimators of the expected value:
 - ▶ sample mean/average: $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$,
 - ▶ median: 50th percentile.
- ▶ Law of iterated (total) expectations: $\mathbb{E}(Y) = \mathbb{E}(\mathbb{E}(Y|X))$.

Variance

- ▶ Measure of variability/dispersion and uncertainty.
- ▶ If $\mathbb{E}(X^2) < +\infty$, then

$$\text{Var}(X) = \mathbb{E}(X - \mathbb{E}(X))^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$$

is called variance of X .

- ▶ Variance is the second centered moment:

$$\text{Var}(X) = \int_{-\infty}^{+\infty} (x - \mathbb{E}(X))^2 f_X(x) dx.$$

- ▶ Standard deviation σ : a positive square root of variance.
- ▶ Some of the properties:
 - ▶ $\text{Var}(X) \geq 0$,
 - ▶ $\text{Var}(b) = 0$, where b is a constant,
 - ▶ $\text{Var}(aX + b) = a^2 \text{Var}(X)$, where a, b are constants.

Covariance

- ▶ Measure of linear association between two random variables.
- ▶ If $\mathbb{E}(X^2) < +\infty$ and $\mathbb{E}(Y^2) < +\infty$, then

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$$

is called covariance between X and Y .

- ▶ Some of the properties:
 - ▶ let $X_1, \dots, X_n$ be random variables with finite second moms. Then

$$\text{Var}(a_1X_1 + \dots + a_nX_n + b) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^n a_i a_j \text{Cov}(X_i, X_j).$$

- ▶ let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be two sets of random variables with finite second moments. Then

$$\text{Cov} \left(\sum_{i=1}^n a_i X_i + b, \sum_{j=1}^m c_j Y_j + d \right) = \sum_{i=1}^n \sum_{j=1}^m a_i c_j \text{Cov}(X_i, Y_j).$$

Correlation coefficient

- ▶ Standardized measure of linear association.
- ▶ Let X and Y be two random variables with finite second moments and non-zero variances. Then

$$\text{Corr}(X, Y) = \rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

is called the coefficient of correlation between X and Y .

- ▶ It holds that (due to the Cauchy-Schwarz inequality):

$$-1 \leq \rho_{X,Y} \leq 1.$$

Uncorrelated random variables

Uncorrelated random variables give some important and practical implications:

- ▶ if $\rho_{X,Y} = \text{Cov}(X, Y) = 0$, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$,
- ▶ if $X_1, \dots, X_n$ are pairwise uncorrelated, then
 - ▶ $\text{Var}(\sum_{i=1}^n X_i) = \sum_{i=1}^n \text{Var}(X_i)$,
 - ▶ $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$.

Outline

Introduction to econometrics

Course information

Quick statistics review

Probability distributions and their features

Recap of selected distributions

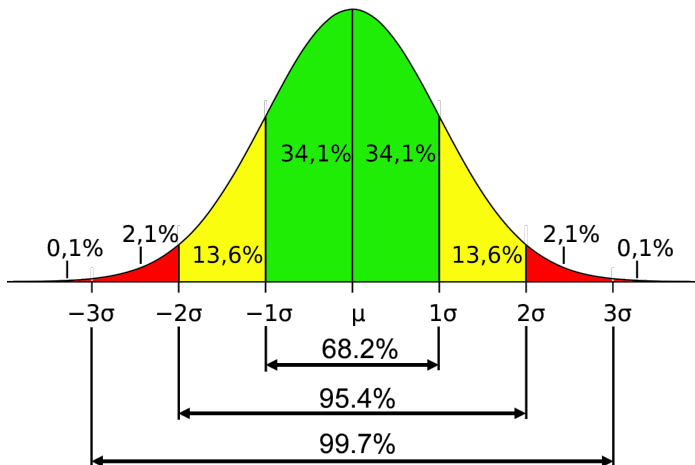
LLN and CLT

Hypothesis testing

Normal distribution

- ▶ Usually referred to as the Gaussian distribution (after Carl Friedrich Gauss).
- ▶ Normally distributed random variable $X \sim N(\mu, \sigma^2)$ has:
 - ▶ $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, x \in \mathbb{R},$
 - ▶ $\mathbb{E}(X) = \mu,$
 - ▶ $\text{Var}(X) = \sigma^2.$
- ▶ $N(0, 1)$ is the standard normal distribution.
- ▶ It is useful to remember the most important quantiles of the standard normal distribution:
 - ▶ $\Phi(1.645) = 0.95,$
 - ▶ $\Phi(1.960) = 0.975$ (≈ 2 , 'two-sigma rule'),
 - ▶ $\Phi(2.326) = 0.99,$
 - ▶ $\Phi(2.576) = 0.995.$

Normal (Gaussian) distribution



Source: www.muelaner.com

Normal (Gaussian) distribution



Source: Wikipedia [here](#)

► Carl Friedrich Gauss

χ^2 distribution

- ▶ If $Z_1, \dots, Z_m$ are *i.i.d.* random variables with $N(0, 1)$, then

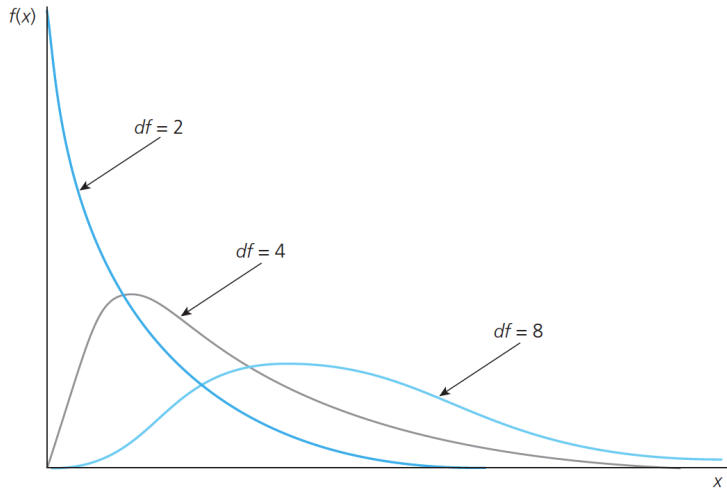
$$X = Z_1^2 + \dots + Z_m^2 \sim \chi_m^2.$$

- ▶ If $X_1, \dots, X_n$ are independent random variables with $\chi_{m_1}^2, \dots, \chi_{m_n}^2$, then

$$Y = X_1 + \dots + X_n \sim \chi_M^2,$$

$$M = \sum_{i=1}^n m_i.$$

χ^2 distribution



Source: Wooldridge (2016, pg. 670)

► χ^2 distribution: R example

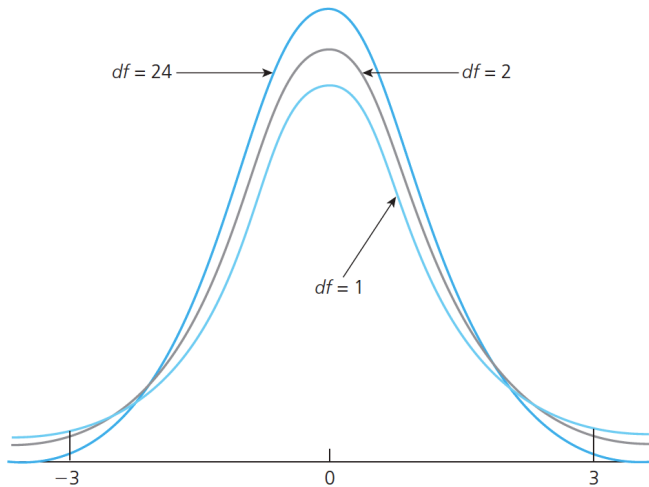
t distribution

- ▶ A.k.a. Student's t distribution (due to William Sealy Gosset's pseudonym).
- ▶ If $Z \sim N(0, 1)$, $X \sim \chi_m^2$, Z and X are independent, then

$$T = \frac{Z}{\sqrt{\frac{X}{m}}} \sim t_m.$$

- ▶ For $m \rightarrow +\infty$, $T \stackrel{as}{\sim} N(0, 1)$.

t distribution



Source: Wooldridge (2016, pg. 671)

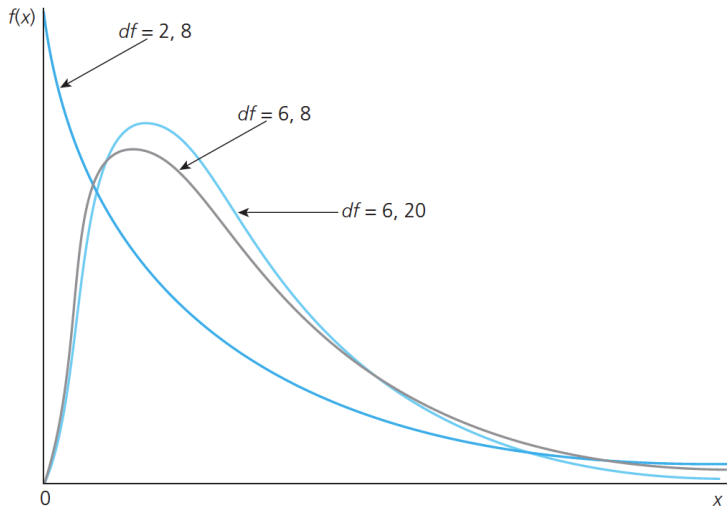
► t distribution: R example

F distribution

- ▶ A.k.a. Fisher-Snedecor distribution.
- ▶ If $X_1 \sim \chi_{m_1}^2$, $X_2 \sim \chi_{m_2}^2$, and X_1 and X_2 are independent, then

$$F = \frac{X_1/m_1}{X_2/m_2} \sim F_{m_1, m_2}.$$

F distribution



Source: Wooldridge (2016, pg. 671)

► F distribution: R example

Outline

Introduction to econometrics

Course information

Quick statistics review

Probability distributions and their features

Recap of selected distributions

LLN and CLT

Hypothesis testing

(Weak) law of large numbers (LLN)

- ▶ Let $X_1, \dots, X_n$ be a sequence of *i.i.d.* random variables with mean μ and finite non-zero variance. Then,

$$\text{plim}_{n \rightarrow +\infty} (\bar{X}_n) = \mu.$$

- ▶ Alternatively written, then for all $\varepsilon > 0$:

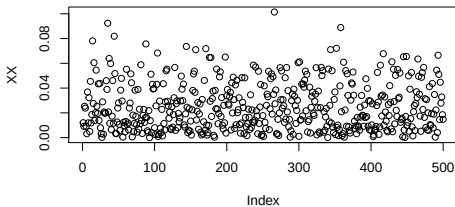
$$\lim_{n \rightarrow +\infty} P(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

- ▶ Intuitively: if we are interested in estimating the population mean μ , we can get arbitrarily close to μ by choosing a sufficiently large sample.

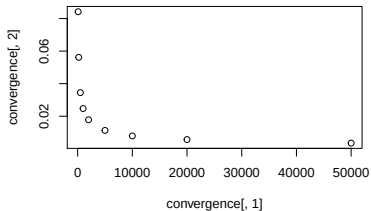
LLN: R example

- ▶ Convergence in probability (with increasing numbers of observations n) of the absolute difference between the sample mean and population mean of *i.i.d.* random variables to zero.
- ▶ Check the 'rate of convergence'.

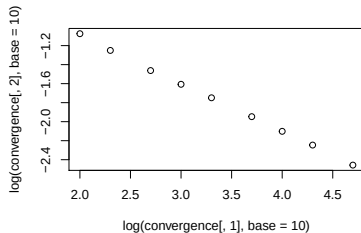
(a) simulated data: $n = 1,000$, $rep = 500$



(b) convergence: $\nearrow n$



(c) log-convergence



Central limit theorem (CLT)

- ▶ Let $\{X_1, X_2, \dots, X_n\}$ be an *i.i.d.* random sample with mean μ and finite non-zero variance σ^2 . Then,

$$Z_n = \frac{\sum_{i=1}^n X_i - n\mu}{\sigma\sqrt{n}} \quad \text{or, alternatively,} \quad Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

has an asymptotic standard normal distribution $N(0, 1)$.

- ▶ Intuitively: regardless of the population distribution of X , Z_n has mean zero and variance one, which coincides with the mean and variance of the standard normal distribution.
- ▶ Remarkably, the entire distribution of Z_n also gets arbitrarily close to the shape of the standard normal distribution $N(0, 1)$ as n gets large.

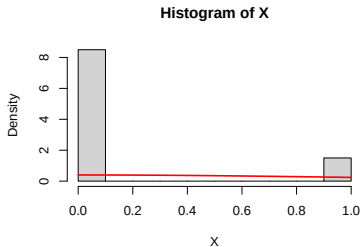
CLT: R example

- ▶ Demonstrate CLT on an extreme case: a binomial distribution.
- ▶ One trial only, probability of success $p = 0.2$:

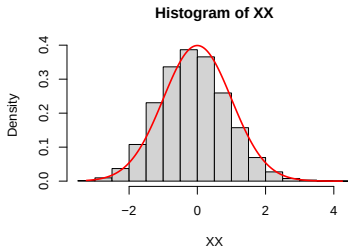
$$\begin{aligned}\mu &= p = 0.2, \\ \sigma^2 &= p(1 - p) = 0.16.\end{aligned}$$

- ▶ Show 'convergence' for different n (an intuitive way of what a 'finite approximation of infinity' could be).

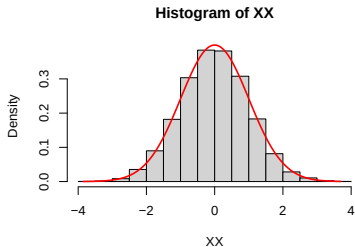
(a) $n = 100, rep = 1$



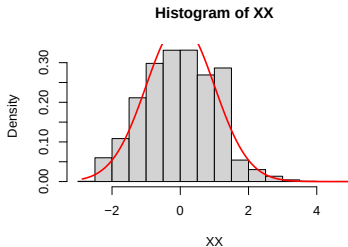
(b) $n = 100, rep = 10,000$



(c) $n = 10,000, rep = 10,000$



(d) $n = 33, rep = 10,000$



Outline

Introduction to econometrics

Course information

Quick statistics review

Probability distributions and their features

Recap of selected distributions

LLN and CLT

Hypothesis testing

Hypothesis testing

- ▶ Null (H_0) and alternative (H_1) hypotheses.
- ▶ Significance level α :
 - ▶ the risk/probability of incorrectly rejecting a true H_0 (Type I error).
- ▶ Test statistic:
 - ▶ function of the random sample, computed for a particular outcome.
- ▶ Critical value(s):
 - ▶ quantile(s) of the distribution of the test statistic under H_0 signifying the range where H_0 is rejected for a given α .
- ▶ p -value:
 - ▶ 'the smallest α at which H_0 is still rejected' for a given test statistic.
- ▶ Type I error:
 - ▶ we reject a true H_0 ,
 - ▶ 'conviction of the innocent' (if H_0 : innocent).
- ▶ Type II error:
 - ▶ we do not reject a false H_0 (usually referred to as β in statistics),
 - ▶ 'letting the criminal go' (if H_0 : innocent).

Hypothesis testing

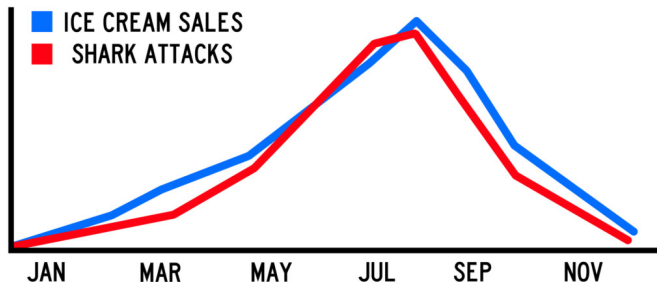
Equivalent approaches (leading to the same results):

- ▶ define H_0 and H_1 ,
 - ▶ set the significance level α , e.g., 5%,
 - ▶ obtain a test statistic,
- a) through the critical value(s):
- ▶ get the critical value(s) of the test statistic under H_0 ,
 - ▶ if the test statistic is outside the quantile range set by the critical value(s), reject H_0 ,
- b) through the p -value:
- ▶ find the complementary CDF probability $P(X > x)$ associated to the quantile equal to the test statistic (p -value),
 - ▶ if the p -value is lower than α , reject H_0 ,
- c) through the confidence interval (usually for two-sided H_1):
- ▶ construct a CI (interval estimate) for a respective parameter,
 - ▶ if the hypothesized value under H_0 is outside the CI, reject H_0 .
- ▶ mind the difference between one-sided and two-sided tests.

Seminars and the next lecture

- ▶ Seminars:
 - ▶ there are no seminars this week.
- ▶ Next lecture #2:
 - ▶ data structures,
 - ▶ simple regression model,
 - ▶ deriving the OLS estimator.
- ▶ Readings for lecture #2:
 - ▶ Chapter 1: 1.3, Appendix C.1,
 - ▶ Chapter 2: 2.1, 2.2, **2.4 (mandatory for/after seminars)**.

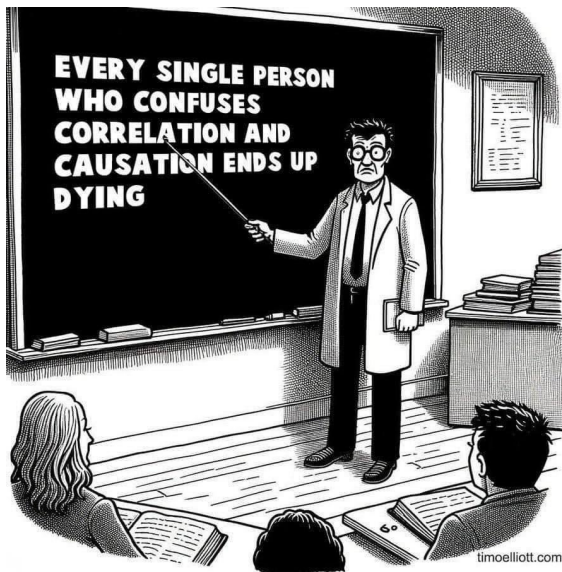
CORRELATION IS NOT CAUSATION!



Both ice cream sales and shark attacks increase when the weather is hot and sunny, but they are not caused by each other (they are caused by good weather, with lots of people at the beach, both eating ice cream and having a swim in the sea)

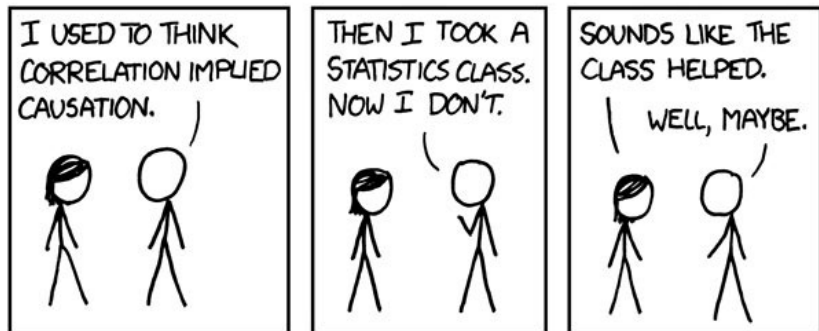
Source: [idatassist.com](https://www.idatassist.com) [2021-10-11]

Appendix I: Causality in econometrics



Source: [X.com](#) [2024-02-18]

Appendix I: Causality in econometrics



Source: xkcd.com/552 [2021-10-11]

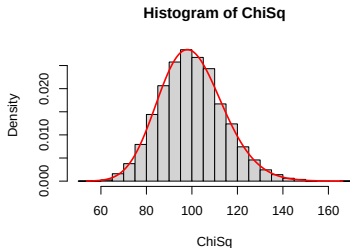
► Back

Appendix II: χ^2 distribution: R example

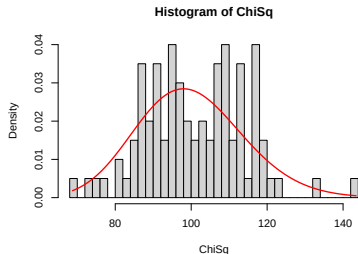
- ▶ How the distribution's shape changes with the df .
- ▶ How close it is to its theoretical density for different numbers of observations/draws.

▶ Back

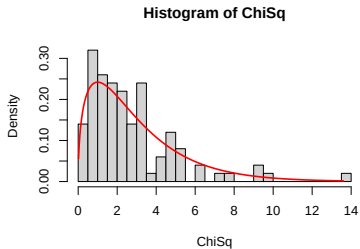
(a) $df = 100, n = 10,000$



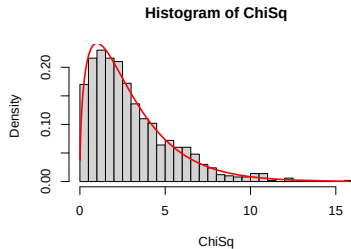
(b) $df = 100, n = 100$



(c) $df = 3, n = 100$



(d) $df = 3, n = 1,000$

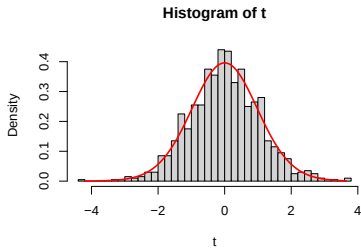


Appendix II: t distribution: R example

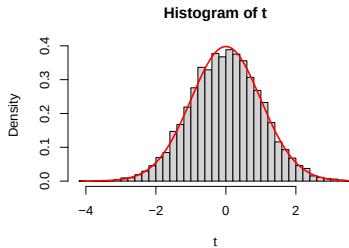
- ▶ How the distribution's shape changes with the df .
- ▶ How close it is to its theoretical density for different numbers of observations/draws.

▶ Back

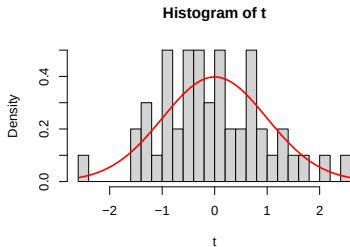
(a) $df = 33, n = 1,000$



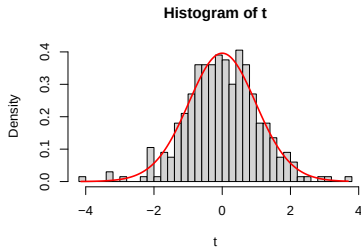
(b) $df = 100, n = 10,000$



(c) $df = 100, n = 50$



(d) $df = 33, n = 333$

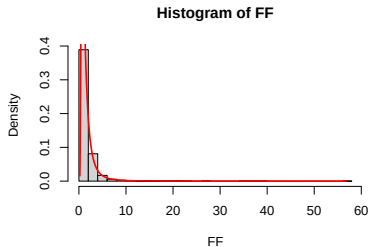
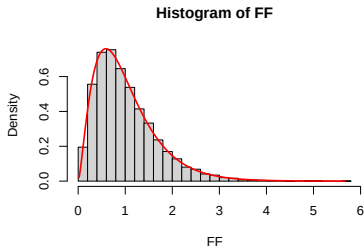


Appendix II: F distribution: R example

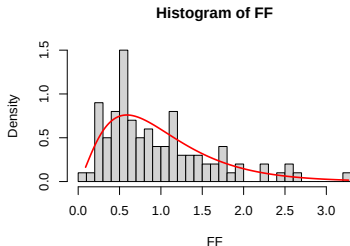
- ▶ How the distribution's shape changes with sets of df , asymmetry.
- ▶ How close it is to its theoretical density for different numbers of observations/draws.

▶ Back

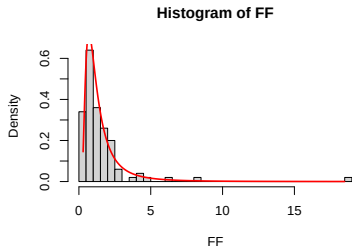
(a) $df1 = 5, df2 = 100, n = 10,000$ (b) $df1 = 100, df2 = 5, n = 10,000$



(c) $df1 = 5, df2 = 100, n = 100$



(d) $df1 = 100, df2 = 5, n = 100$



Historical spotlight: Carl Friedrich Gauss (1777–1855)

Carl Friedrich Gauss, often called the 'Prince of Mathematicians,' is why we say 'Gaussian.' He was born in Brunswick, Germany, and spent almost his entire career at the University of Göttingen, where he also ran the Göttingen Observatory.

He shaped the normal curve and used the method of least squares to recover the orbit of Ceres in 1801, at age twenty-four. Ceres was a new object between Mars and Jupiter that astronomers briefly lost. With only a few noisy measurements, Gauss predicted where to look next. They aimed their telescopes there and found it again.

The famous classroom legend goes like this: the teacher wanted quiet time, so he gave the class a 'never-ending' task: add 1 to 1000. Pencils down, brains off. Young Gauss looked once and thought, this is easy. He paired 1 with 1000, 2 with 999, 3 with 998. Each pair makes 1001. There are 500 pairs, so $500 \times 1001 = 500,500$. He wrote the answer, closed his chalkboard, and sat there smiling while everyone else kept counting forever. The teacher's plan for peace and quiet was ruined by one very smart ten-year-old. Historians note that versions of the story differ, but the lesson for econometrics is simple: look for structure, use smart algebra, and avoid brute force.



Source: [Wikipedia](#)